



— CAPABILITY STATEMENT

SCIENCE SYSTEMS AND APPLICATIONS, INC.

SCIENCE AND TECHNOLOGY WITH PASSION

UEI PJ2SP76D3HJ3

CAGE 5S009

NAICS 541715 (small) / 541330

WOSB · ITAR Registered

ISO 9001 · AS9100 · CMMI L3

Top Secret Facility Clearance

We turn science into mission advantage. SSAI delivers mission-critical science, engineering, and technology to the agencies that explore, observe, and defend. For more than four decades, we have turned complex science into systems that perform when the mission cannot fail, from NASA and NOAA to defense and national security.

1977

FOUNDED · 49+ YEARS

350+

SCIENTISTS &
ENGINEERS

160+

MISSIONS SERVED

100+

MAJOR INSTRUMENTS,
MODELS & MISSIONS

10,000+

PUBLICATIONS & DATA
PRODUCTS

3

WHOLLY-OWNED
SUBSIDIARIES

CORE CAPABILITIES

THE SCIENCE AND ENGINEERING BEHIND THE MISSION.

We serve the scientists, engineers, forecasters, analysts, and operators who explore, observe, and defend. Five capabilities carry the mission from research and instrument engineering to manufacturing, environmental intelligence, and secure digital defense.

- 01**
SCIENCE, TECHNOLOGY & ENGINEERING SERVICES
We engineer the instruments and systems that have to work the first time, on missions that cannot fail. The disciplines behind that outcome: systems engineering, requirements and V&V, avionics design, software engineering (CMMI L3), FPGA design, EEE parts processing, system integration and testing, and manufacturing engineering, to AS9100, ITAR, and J-STD-001F workmanship.
- 02**
SPACE OPERATIONS
We keep the nation watching the planet and the space around it, every day the mission demands. We operate NASA's GEOS Data Assimilation System (millions of observations daily), global atmospheric reanalysis, and 24/7 operational modeling for NOAA and NASA, plus deep-space instrument development and calibration, trajectory design, and mission operations from pre-launch through on-orbit.
- 03**
SATELLITE & SPACE SYSTEMS MANUFACTURING
We build flight hardware engineered to work the first time, and every time. A 32,000 sq ft electronics manufacturing facility and a 10,000 sq ft machine shop under two miles from NASA Goddard, with an ESD-protected floor, three clean rooms, cleared personnel, Mil-spec CNC machining, and certified aerospace machinists, end to end from prototyping to production.
- 04**
ENVIRONMENTAL INTELLIGENCE
We turn what sensors see into decisions the mission can act on. Data assimilation, remote sensing, oceanography, hydrology, biosphere modeling, and weather and space-weather forecasting, across instruments including OMI, OMPS, MODIS, VIIRS, MISR, ICESat-2, AVIRIS-NG, TROPOMI, and GEMS.
- 05**
CYBERSECURITY & DIGITAL DEFENSE
We make platforms smarter and secure, so the mission stays ahead. Advanced IT, cybersecurity on zero-trust architectures, and AI/ML for digital twins, anomaly detection, and cross-agency support, anchored by CREST (Coupled Reusable Earth System Tensor), an AI-enabled Python digital-twin framework. Engineered for national security, designed to work every time.

DIFFERENTIATORS

A PARTNER THE MISSION CAN TRUST.

- **Beside NASA Goddard, in the D.C. defense corridor.** We work where the missions are engineered and flown, minutes from NASA Goddard and the Washington defense community.
- **Trusted for more than four decades.** Founded in 1977, we field 350+ Scientists and engineers who have delivered across 160+ missions.
- **One trusted source for the whole mission.** Three wholly-owned subsidiaries bring electronics manufacturing, cybersecurity engineering, and precision machining in house.
- **Cleared and certified for the hardest work.** We hold a Top Secret Facility Clearance and ITAR registration, with appraised process maturity and aerospace-grade quality systems behind every delivery.

AS9100 Aerospace Quality

CMMI L3 Development

ISO 9001:2015

Top Secret FCL

ITAR Registered

SUBSIDIARIES

MISSION CAPABILITY, IN HOUSE.

Three wholly-owned subsidiaries put the capability the mission needs under one roof.

ADVANCED MISSION PARTNERS

Electronics manufacturing, engineering services, EEE expertise, and precision manufacturing.

ELUCIDATION CONCEPTS, LLC

Cybersecurity solutions, systems engineering, system integration testing, and test & evaluation.

SIGMA ENGINEERING & FABRICATION, INC.

Precision machining, manufacturing, material expertise, and quality at speed.

PAST PERFORMANCE

TRUSTED ON THE MISSIONS THAT CANNOT FAIL.

ESES III	NASA Goddard, Engineering Technology Directorate (CPAF). Study, design, development, fabrication, integration, testing, verification, and operations of spaceflight, airborne, and ground hardware and software. 11/2018 to 04/2025.
HBG	Hydrospheres, Biospheres and Geophysics. NASA Goddard, Earth Sciences Division (CPFF). Scientific research services across space, airborne, in-situ, and laboratory observations. 04/2020 to 11/2025.
STARSS III	Science, Technology and Research Support Services. NASA Langley, Science Directorate (CPFF). Earth atmosphere and climate, extending to other planetary atmospheres. 12/2016 to 05/2023.
SAMDA	Support for Atmospheric Modeling and Data Assimilation. NASA Goddard, Global Modeling and Assimilation Office (CPFF). Research across the Earth Sciences Division. 03/2017 to 11/2025.
HBS	Hydrospheric and Biospheric Sciences. NASA Goddard, Earth Sciences Division (CPFF). 12/2014 to 05/2020.

Flagship missions: OCI / PACE · Roman Space Telescope · Lucy · Mars Sample Return / CCRS · Dragonfly · GeoXO · DAVINCI+ · TSIS-2 · JWST.

MARKETS SERVED

TRUSTED ACROSS SIX MISSION COMMUNITIES.

DEFENSE & NATIONAL SECURITY

DoD and defense labs.

FEDERAL CIVILIAN

NASA, NOAA, USGS, EPA, USDA, GSA, Commerce, Interior.

INTELLIGENCE COMMUNITY

Secure mission support.

STATE & LOCAL

LA World Airports, LA Unified School District.

COMMERCIAL

Industry and partners.

OCONUS / INTERNATIONAL

World Bank and global programs.

Customer set: NASA (Goddard, Langley), NOAA, USGS, EPA, USDA, GSA, Department of Commerce, Department of the Interior, DoD and defense labs, LAWA, LAUSD, and the World Bank.

CORPORATE DATA

THE ESSENTIALS.

UEI	PJ2SP76D3HJ3	CONFIRM	CAGE	5S009	CONFIRM
NAICS	541715 (small) / 541330		BUSINESS STATUS	WOSB, SBA 2024	CONFIRM
GSA MAS	47QRAA23D0014	CONFIRM	IDIQ VEHICLE	OASIS Small Business	CONFIRM
QUALITY & MATURITY	ISO 9001:2015, AS9100, CMMI L3		SECURITY	Top Secret FCL, ITAR Registered	
FOUNDED	1977		HEADQUARTERS	Lanham, MD	

BUSINESS DEVELOPMENT

BD@ssaihq.com
(301) 867-2000

HEADQUARTERS

The Aerospace Building
10210 Greenbelt Rd, Suite 600
Lanham, MD 20706

LOCATION ADVANTAGE

Minutes from NASA Goddard, in the Washington D.C. defense corridor.

Procurement identifiers and contract values marked "confirm" / "SSAI to confirm" are pending SSAI sign-off before external release.